

[Gerald. "Wrongful convictions and DNA exonerations: Understanding the role of forensic science." *NIJ Journal* 279 (2017): 250705.]

Unpredictable juror decision-making, the unknown impact of other contributing factors, subjective assessments of information, and lack of complete information result in some uncertainty in how much forensic science has contributed to wrongful convictions.

However, there are some critical lessons that forensic scientists can take away from these findings. First, forensic misconduct is fervently unacceptable; it has a pervasive and infectious effect on the entire criminal justice system. All forensic laboratories and forensic scientists are obliged to make every effort to prevent forensic misconduct. Accreditation, implementation and enforcement of a code of ethics, and appropriate training should mitigate forensic misconduct. Second,

forensic scientists must avoid ambiguous terminology in their reports and testimony because they will mislead investigators, litigators, and factfinders.

Forensic science professionals strive to convey their findings accurately and reliably. **But terminology** such as "consistent with," "similar to," and "cannot be differentiated" — qualitative terms that forensic

scientists often use to avoid making conclusive statements that two or more items are not from the same source — **may be interpreted differently by courts and juries when used in a certain context and not fully explained. These**

ambiguous phrases can have repercussions beyond what they were originally intended to do, which is for the forensic scientist to communicate uncertainty. Therefore, all forensic disciplines need to clearly define the language they will use and be cognizant of potential misinterpretation by nonscientists. Third, and along related lines, forensic scientists must convey impartial and objective conclusions based on accurate and reliable techniques. It is also just as important to clearly articulate limitations and uncertainty so that all users understand the confines of the forensic findings. In reviewing the erroneous convictions that involved forensic serology, there appears to be an underlying issue with mixture interpretation and statistical assessments. Forensic scientists need to demonstrate core competency in the use of and interpretation of statistics. Having an advanced level of statistical training through undergraduate and graduate forensic science programs is essential. Fourth, errors are often inevitable; when they do occur, it is critical to focus on the underlying problems that contributed to the event — and then to learn from the error. This is an especially challenging issue because the general tendency is to blame an individual. The final recommendation is best stated in a quote from Dr. Paul Camille Hippolyte Brouardel, a French pathologist: "If the law has made you a witness, remain a man of science. You have no victim to avenge, no guilty or innocent person to convict or save — you must bear testimony within the limits of science."¹⁸